

MUXI.AI

Multiplexed, eXtensible Intelligence



A modular, open framework and cloud platform to
safely build, deploy, and monetize AI agents

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EXECUTIVE BRIEF

A strategic summary of MUXI's vision, innovation, and opportunity.

MUXI: The Infrastructure Standard for an Agentic Future

The Docker for AI agents – open, compliant, and ready for Europe's AI future

Highlights:

- MUXI is the first infrastructure that makes AI agents as easy to deploy as websites
- We're not building another AI framework. We're building the Docker for AI Agents.
- *While everyone else builds agents, we're building the infrastructure that makes them actually work.*

Executive Summary

What is MUXI?

MUXI (Multiplexed, eXtensible Intelligence) is the first **AI Infrastructure Layer** for deploying specialist AI agents. Think of it as the **"Docker for AI Agents"** – creating the same standardization and portability revolution for AI systems that Docker created for applications.

The Problem We Solve

Currently, deploying AI agents requires stitching together 12+ different tools manually. MUXI changes this by providing a single, self-hostable runtime that handles all the complexity behind the scenes.

What Makes MUXI Different

MUXI is not a chatbot wrapper or agent framework. It's an **intelligent infrastructure layer** that:

- Runs like enterprise-grade infrastructure
- Thinks and orchestrates like a sophisticated AI partner
- Handles multi-agent coordination, memory management, and document processing natively
- Deploys anywhere from laptops to data centers

The Vision

Just as web servers power websites and container runtimes power microservices, MUXI powers intelligent AI systems. We're building the foundation for a new era of open, standardized AI tooling that developers can trust and control.

🔥 What Makes MUXI Revolutionary:

- 🧠 **True AI Autonomy:** Tell it what you want, not how to do it – unlike every workflow-dependent competitor, MUXI understands, breaks down, and executes complex tasks without needing workflow blueprints
- 🎯 **The Overlord Advantage:** One conversation interface that secretly coordinates armies of specialist agents behind the scenes while maintaining natural, human-like interactions
- 📚 **Memory That Actually Works:** Learns what matters, forgets what doesn't – like having a digital brain that gets smarter over time
- 🤝 **Agent Teamwork:** Native local and remote agent-to-agent communication that just works out of the box
- 📄 **Document Intelligence:** Handles PDF, Word, and image documents natively – with built-in OCR and semantic understanding
- ⚡ **Deploy Like Magic:** `muxi push` and you're live – as simple as deploying a website with Vercel

We are seeking support from the European Innovation Council (EIC) to complete the development of MUXI's advanced orchestration capabilities, expand its intelligent infrastructure features, and establish Europe's leadership in open, ethical, and deployable conversational AI infrastructure.

🚀 Summary of Innovation:

MUXI introduces the world's first open, declarative, and infrastructure-native runtime for deploying AI agents at scale. Unlike frameworks that embed orchestration within application code, MUXI externalizes this logic into a self-hostable AI server, enabling version-controlled agents, runtime observability, and secure, multi-agent collaboration – all deployable via Docker on local or cloud environments.

This architectural shift enables developer ownership, transparency, and EU-grade compliance in a space currently dominated by proprietary SaaS platforms.

The AI Infrastructure Problem

The AI industry has a dirty secret: While everyone's building agents, no one can actually deploy them reliably.

Developers are currently struggling with:

- **Frankenstein integrations** of 12+ different tools just to get basic agents running
- **API call complexity** that breaks every time a provider changes something
- **Zero observability** into what their AI is actually thinking or doing
- **Vendor lock-in concerns** with closed-source platforms
- **Infrastructure complexity** where deploying one agent requires managing multiple services

Most current approaches are ad hoc – developers stitch together memory, tool use, LLM prompts, and orchestration logic manually. These systems are hard to deploy, scale, or monitor, and often rely on opaque, closed-source APIs.

Meanwhile, Big Tech is making their move. OpenAI Assistants, Google Vertex AI, and Microsoft Copilot Studio are all racing to lock up this space with proprietary platforms.

Without open infrastructure, European developers and enterprises will be permanently locked out.

What The Market Needs

- **Infrastructure teams:** Observability, control, and portability – not another black box
- **Enterprise AI teams:** Privacy and compliance – not vendor surveillance
- **Developers:** To run agents like servers or daemons – not mess around with prompt files in Jupyter
- **European businesses:** Infrastructure they can trust and control

MUXI solves all of this by introducing a new abstraction layer: The AI Server.

Solution: MUXI – The AI Infrastructure Layer

MUXI introduces the first developer-first, conversational, self-hostable runtime for intelligent AI systems.

This addresses the fundamental gap in AI infrastructure that the industry has been waiting for.

Core Intelligence Capabilities:

- **Overlord Orchestration:** Advanced conversation management that maintains natural interactions while coordinating specialist agents behind the scenes
- **Async Request-Response System:** Intelligent handling of long-running agentic tasks with threshold-based async processing, webhook integration, and comprehensive request tracking
- **Document Intelligence:** Native processing of PDFs, Word documents, and images with OCR built-in – no manual setup required
- **Autonomous Intelligence:** Tell it what you want, not how to do it – natural task understanding without predefined workflows
- **MCP Tool Integration:** Native access to hundreds of thousands of tools through standardized interfaces
- **Agent-to-Agent Communication:** Sophisticated agent collaboration with no manual coordination needed
- **Smart Memory Management:** Intelligent memory that learns what's important and forgets what's not – optimizing focus over time

Enterprise Infrastructure Features:

- **Built-in Observability:** Complete visibility into agent thinking, actions, and decisions – no more black box operations
- **Traceable Chain-of-Thought:** Full audit trails of agent reasoning – transparency and safety built-in
- **Secure Code Execution Sandboxing:** Safe remote code execution with shared containers for stateless tasks and persistent sandboxes for autonomous development environments
- **Infrastructure-as-Code:** Define agents with YAML configs – version control, rollbacks, and reproducibility included
- **Docker-Native Deployment:** Ships with local LLMs bundled – deploy anywhere from laptop to data center
- **Multi-LLM Support:** Local, hosted, or hybrid – intelligent routing picks the optimal model for each task
- **Complete DevTools:** Web dashboard, CLI, and SDKs for Python, JS, Go – comprehensive development experience

MUXI lets developers deploy specialist AI agents like microservices – conversational, intelligent, observable – but with enterprise-grade reliability.

Technical Architecture

MUXI is composed of four major layers, mirroring the Docker ecosystem:

1. MUXI Server: The agent runtime engine

Analogy: Docker Runtime

- Licensed under Elastic 2.0
- Manages agent lifecycles, tool interaction, and task routing
- Runs local LLMs or calls external APIs
- **Advanced Async Orchestration:** Handles long-running agentic tasks with intelligent threshold-based processing and webhook integration
- **Integrated Sandboxing:** Secure code execution via mandatory RCE service for stateless tasks and persistent containers for autonomous agents
- Self-hostable, Docker-deployable infrastructure layer

2. MUXI Formation: Agent configuration containers

Analogy: Docker Containers

- Formation-based YAML configurations that package AI agents as portable, reproducible units
- Version-controlled agent definitions with dependencies, memory, and tool specifications
- Multi-agent orchestration: Task delegation and coordination are handled via a built-in Overlord engine.

3. DevTools: Developer ecosystem

- **CLI:** Command-line interface for agent management and deployment
- **SDKs:** Python, JavaScript, Go, and other language bindings
- **Studio:** Web-based UI for agent development, testing, and monitoring

4. Open Registry: Agent and Tool marketplace

Analogy: Docker Hub

- Public registry for sharing and discovering agents and tools
- Developers can share their own agents and tools with the community
- Marketplace for both free and commercial agents, MCP tools, and formations

Revenue Model: MUXI Cloud

Analogy: Vercel for AI Agents

MUXI Cloud provides hosted infrastructure and marketplace services, creating sustainable revenue streams while maintaining the open-source foundation:

Hosting Revenue Streams:

- **Flex Plan:** \$20/month when active + VPS pricing (pay-as-you-go)
- **Pro Plan:** \$10/month annually + VPS pricing (persistent deployment)
- **Pro + Analytics Bundle:** \$20/month annually + VPS pricing (includes observability)
- **Team Plans:** \$49/user/month (capped at \$499/month) with collaboration features
- **Startup Plans:** \$99/month for early-stage companies (<\$1M funding)
- **Enterprise:** Custom pricing for compliance and private cloud deployments

Marketplace Revenue Streams:

- **Agent Marketplace:** Revenue sharing from commercial agents and formations
- **MCP Tool Marketplace:** Monetization platform for Model Control Protocol services
- **Professional Services:** Custom agent development, integrations, and consulting
- **Add-on Services:** Trail Analytics, hosted models, horizontal scaling, private endpoints

Revenue Projections:

Based on market analysis and conversion modeling, MUXI Cloud demonstrates strong revenue potential:

- **Conservative (3% conversion):** €3.6M ARR at 100K users
- **Moderate (5% conversion):** €6.0M ARR at 100K users
- **Optimistic (10% conversion):** €12.0M ARR at 100K users

The marketplace model creates network effects where both agent creators and consumers benefit, establishing MUXI as the central ecosystem for AI agent monetization and distribution.

Differentiators

Key Differentiators:

- **Truly Autonomous:** MUXI agents think for themselves – understanding and breaking down tasks without requiring predefined workflows, unlike workflow-dependent platforms
- **MUXI is a server**, not a library or SDK. It runs agents like a process manager runs daemons.
- **Declarative-first:** Agents are defined via configuration, enabling version control, portability, and predictable behavior.
- **Multi-agent orchestration:** Task delegation and coordination are handled via a built-in Overlord engine.
- **Language-agnostic integration:** Interact via SDKs or REST/HTTP, not embedded Python code.
- **Privacy and compliance-ready:** Deploy entirely locally or hybrid, with built-in observability.

MUXI turns agent orchestration into infrastructure, not application code. Unlike other frameworks that offer components for building agents, MUXI provides a complete, opinionated agent architecture with built-in orchestration, memory, and communication.

The comprehensive ecosystem approach addresses current AI development fragmentation where developers are forced to cobble together scattered, incompatible tools and API calls. No other project currently matches its combination of portability, structure, tooling, and open extensibility.

Unlike Langchain or CrewAI, which focus on agent composition within Python applications, MUXI offers a **runtime-first, infrastructure-native approach** – turning agent deployment into an operational capability, not an embedded script. MUXI treats agents as first-class deployables, enabling true observability, versioning, and orchestration at the system level.

Comparison Table

MUXI occupies a unique technical and strategic niche that is not served by today's leading agent frameworks or orchestration tools:

Feature	MUXI	Crew.ai	Langchain	Agno Agent API
Product Type	AI Infrastructure Layer	Agent-as-a-Service	Python Framework	Developer API Framework

Autonomous Task Handling	✅ Completely autonomous – no workflows needed	❌ No	❌ Workflow-dependent	❌ Workflow-dependent
Self-hostable	✅ Full Docker support	❌ Cloud-only	⚠️ Not runtime-native	✅ Requires FastAPI setup
Conversational AI Orchestration	✅ Overlord with intelligent coordination	❌ No	❌ No	❌ No
Document Understanding & OCR	✅ Native PDF/DOCX/Image processing	❌ No	⚠️ Manual integration	❌ No
Smart Memory Management	✅ Built-in intelligent memory	❌ No	⚠️ Manual setup	⚠️ Basic vector support
A2A Protocol & Cross-Formation	✅ Native agent communication	❌ No	❌ No	❌ No
MCP Integration	✅ Native first-class citizen	❌ No	❌ No	⚠️ Basic tool support
Multi-User Support	✅ Native enterprise-grade design	❌ No	⚠️ Custom implementation	❌ No
WebRTC & Real-time Streaming	✅ Native multi-modal streaming	❌ No	❌ No	❌ No

Infrastructure-as-Code	✓ Formation YAML configuration	✗ No	✗ No	⚠ Python-only
Multi-agent orchestration	✓ Seamless behind-the-scenes routing	✓ Yes	⚠ Manual setup	✗ Single-agent only
Authentication & Access Control	✓ Comprehensive built-in system	✗ No	⚠ Basic auth	⚠ Custom implementation
Chain-of-Thought Tracing	✓ Full reasoning transparency	✗ Basic	⚠ Limited	⚠ Minimal logging
Async Long-Running Tasks	✓ Native threshold-based async handling	✗ No	✗ No	✗ No
Secure Code Execution	✓ Built-in sandboxed remote code execution (RCE) + persistent containers	✗ No	✗ Manual integration	✗ No
Cloud & local parity	✓ MUXI Cloud (opt-in)	✗ SaaS only	✗ No	✗ No

Development Status & Readiness

MUXI is ~70% feature-complete, with the core runtime already operational, deployable, and in use by early testers. This proposal accelerates the transition from stable MVP to a polished, production-ready platform.

The core MUXI runtime is fully operational with 22 major components implemented, and already demonstrates:

- **Sub-second response times** for agent interactions
- **Multi-formation deployment** with A2A cross-communication
- **Production-ready** Docker containers with local LLM support
- **Enterprise security** with encrypted secrets and comprehensive authentication
- **Developer-friendly** CLI tools and SDK integration

⚡ This is strategic acceleration funding, not basic research.

Target Users & Beneficiaries

- AI/ML Engineers
- DevOps and infrastructure teams
- Indie AI builders and open-source devs
- Enterprises looking to build agent-powered internal tools with compliance
- National and academic institutions focused on ethical AI

Use of Funds

We are seeking €1.2M to support:

- **Core development** of the MUXI runtime and orchestration engine (mostly done)
- **Developer Relations & Community** (events, education, early access programs, workshops, developer advocacy) (40%)
- **SDKs and DevX (docs, examples, integrations)** (15%)
- **Cloud dashboard and scaling tooling** (35%)
- **Legal, security, licensing, and EU-standard compliance** (10%)

We will use the funding to expand our core engineering team, with a focus on AI infrastructure specialists and EU-based DevRel hires to support workshops, hackathons, and partner onboarding across key European hubs.

Milestones

Phase	Timeline	Deliverables
MVP Runtime	Q2, 2025	Dockerized YAML-first AI Server, local LLMs, CLI agent runner

Partner SDKs	Q3, 2025	Python/TS SDKs, schema validators, integration tooling
MUXI Cloud Alpha	Q1, 2026	Hosted dashboard, Overlord monitoring, remote deploy interface
Enterprise Toolkit	Q2, 2026	SSO, usage dashboards, commercial licensing, support plans

Ecosystem Strategy & Business Sustainability

MUXI creates a complete ecosystem including AI Server, CLI, SDKs, Registry, observability tools, Studio, and Cloud services – establishing a platform developers can grow into rather than outgrow. This comprehensive approach creates multiple value propositions and potential revenue streams while building a sustainable foundation for long-term growth.

For example, a small EU fintech can use MUXI to deploy a secure, multilingual support agent that routes complex requests to compliance experts, automates onboarding, and adapts to local regulations – all without needing to rely on OpenAI or outsource core data processing to non-EU clouds.

Developer Relations Strategy:

A significant portion of EIC funding (40%) is allocated to developer relations and community building, recognizing that platform success depends on developer adoption. This includes:

- **Developer advocacy** and thought leadership in the AI agent space
- **Educational content** including tutorials, best practices, and use case documentation
- **Community events** including conferences, meetups, and hackathons across Europe
- **Early access programs** for enterprise partners and open-source contributors
- **Technical workshops** and certification programs for AI agent development

Revenue Sustainability:

Financial projections demonstrate potential for €3.6M-€12M ARR at 100K users with 3-10% conversion rates, showing long-term viability without dependence on grant funding. The developer-first design emphasizes observability, version control, and CLI-first workflows, demonstrating deep understanding of developer needs and production requirements.

Strategic Risk Mitigation:

- **Market timing risk** mitigated by growing multi-agent adoption across enterprises
- **Complexity vs. adoption** balanced through simple common cases while keeping complex scenarios possible
- **Standardization dependency** addressed through MCP protocol adoption and Formation YAML format gaining broader acceptance

While multi-agent maturity is still evolving, demand for orchestration, observability, and privacy is rising fast – and these infrastructure challenges are orthogonal to LLM capabilities. MUXI is designed to grow with the ecosystem, offering value now and a stable foundation for future agent architectures.

Long-Term Vision

MUXI isn't just infrastructure – it's the foundation of the Agent Economy.

We're creating the "**App Store for AI Agents**" – but unlike Apple's walled garden, ours is open, self-hostable, and owned by developers.

The Agent Economy

By introducing Formation-based agent packaging and a runtime that supports both local and cloud execution, MUXI creates the foundation for a global marketplace of interoperable agents.

Imagine a world where AI agents are as easy to find, deploy, and monetize as mobile apps:

- **Publish:** `muxi push` your agent and instantly reach millions of developers
- **Discover:** Browse thousands of battle-tested specialist agents for any use case
- **Deploy:** `muxi pull` and deploy agents across environments in seconds
- **Monetize:** Earn revenue from agent usage with built-in payment infrastructure

Network Effects & Ecosystem Growth

The marketplace model creates powerful network effects where:

- * **Agent creators** gain access to a global distribution platform
- * **Developers** benefit from a vast ecosystem of ready-made specialist agents
- * **Enterprises** can build on proven, community-tested agent foundations
- * **MCP tool developers** monetize specialized services and integrations

Cloud-Native Developer Experience

MUXI Cloud combines the operational simplicity of Vercel with the power of Docker Hub, offering seamless workflows for deploying and managing intelligent agents across local and cloud environments:

- `agent pull`, `agent push`, and `agent deploy` commands for instant deployment
- Managed infrastructure without sacrificing transparency or portability
- Self-hostable core ensures developers maintain control and avoid vendor lock-in

MUXI becomes the standard that everyone builds on.

Just like:

- **Nginx** became the default web server
- **Docker** became the default container runtime
- **Kubernetes** became the default orchestrator

MUXI will be the default agent runtime – open-source, locally runnable, extensible, and ready for EU-grade compliance.

The EU has a choice: Lead the AI infrastructure revolution with transparent, developer-owned systems – or watch Silicon Valley lock up another entire technology generation.

MUXI is Europe's answer.

Licensing & Openness

MUXI is:

- Licensed under **Elastic License 2.0** for core runtime
- Apache 2.0 for schema, SDKs, CLI, and developer tools
- Structured to prevent hosting abuse while enabling community growth

We intend to apply for IP Box treatment in Cyprus and maintain a dual-path model: open source + commercial cloud.

MUXI is fully compatible with Docker licensing and container runtime norms, ensuring frictionless deployment without licensing entanglements.

Team & Execution

Led by Ran Aroussi, a 30+ year tech veteran and founder of Automaze, MUXI's team combines:

- Deep backend and cloud architecture experience
- Long-standing open-source contributions (e.g., yfinance w/ 3M downloads/month)
- A track record of indie execution, product-market fit, and platform building
- Strong developer relations and community building experience from open-source ecosystem engagement

We're already building in public and onboarding alpha testers for private deployments, with active developer community engagement through technical content, conferences, and direct feedback channels.

Why the EIC

By offering a fully self-hostable infrastructure layer, MUXI supports Europe's strategic autonomy in AI – ensuring that enterprises, governments, and developers are not dependent on foreign cloud providers or opaque systems. MUXI aligns with Europe's Digital Decade goals and the EU AI Act by enabling privacy-first, explainable AI tooling that is developer-owned and locally deployable.

The EIC grant enables us to:

- Accelerate infrastructure tooling without VC pressure
- Maintain open governance and EU-first standards
- Grow MUXI as a reference project for open AI in Europe

European strategic impact:

- Sovereignty: Enables EU enterprises and governments to deploy agent infrastructure without reliance on foreign APIs or clouds.
- Compliance-first by design: Local LLMs, agent observability, and audit trails support GDPR and the EU AI Act.
- Developer empowerment: Provides European developers with infrastructure-grade tools comparable to Silicon Valley's best, but with open governance.
- Fosters innovation: Establishes the EU as a leader in agent-native tooling, paving the way for agent-based startups across finance, health, and govtech.

We are committed to keeping MUXI open, developer-first, and **hostable anywhere – not locked in Silicon Valley's data centers**.

Contact: Ran Aroussi - ran@aroussi.com, [@aroussi](https://twitter.com/aroussi)

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What is MUXI?

MUXI is a modular, open-source platform for building, deploying, and monetising intelligent agents. It helps developers turn AI workflows into scalable, interpretable, and monetisable systems – fast. From secure tooling to hosted infrastructure and a developer marketplace, MUXI is the foundational layer for the agent-powered future.

Why now?

AI agents are no longer a concept – they're users. They navigate interfaces, trigger workflows, and complete tasks across the web. But current systems weren't built for them.

Just like the early internet evolved for mobile, we're now entering the **AI experience (AX)** era – where UX must accommodate both humans and machines.

MUXI is the infrastructure layer for this transition. It's not just about enabling AI – it's about making AI **usable, observable, and scalable**. From developer-first tooling to secure agent orchestration and

monetisation, MUXI helps businesses and developers adapt to the structural shift happening in software.

Core components

- **Framework:** Declarative agent architecture (open-source)
- **Tooling:** Modular backend tools via MCP (search, calendar, email, etc.)
- **MUXI Cloud:** Vercel-style agent deployment + observability
- **Marketplace:** Monetisation and reuse for agent developers

What makes it innovative?

- Declarative-first AI architecture
- Runtime chain-of-thought + memory logs
- Inter-agent communication (local + remote)
- Multiplexed runtime for multimodal data
- Business layer: cloud deployment + “app store” for AI agents

Market opportunity

- Projected ARR of \$50–100M within 24 months
- Developer-first model = scalable adoption
- Monetisation built in for both agents and hosted services

Licensing

Open-source with restrictions on competing cloud platforms.

Funding use

Grant covers:

- Open-source and community development
- Hosted tooling for agent builders
- Scalable infrastructure for cloud deployment

Marketplace + enterprise growth = self-funded.

Strategic relevance

MUXI is developed and maintained by **Automaze Ltd**, a UK-based software development agency with extensive experience in building scalable infrastructure and AI-driven applications. Automaze provides the technical delivery team, ensuring UK-based IP creation and product reliability.

- UK-based IP creation and technical leadership
- AI adoption in public and private sectors
- Infrastructure for safe, traceable, and modular AI usage

MUXI isn't just another AI tool. It's the infrastructure to power the agent economy.

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BUSINESS PLAN

** All cost and revenue projections are noted in USD
due to the global nature of the business.*

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1. Executive summary

MUXI is an open-source platform for building, deploying, and monetising intelligent agents. As the software industry enters the AI experience (AX) era, where systems must accommodate both human and machine users, MUXI provides the infrastructure needed to power this shift.

Developed by Automaze Ltd (UK), MUXI lowers the barrier for AI adoption across public and private sectors by offering secure, interpretable, and scalable tooling for agent developers. Our goal is to create a new layer of developer infrastructure – like a Vercel or Heroku for AI agents – complete with open protocols, hosted tooling, and monetisation pathways.

2. Vision and mission

Our vision is to create a world where agents become a first-class layer of the internet – replacing manual software with intelligent, reusable systems that act on behalf of users and organisations.

MUXI is the foundation for this transition: a modular agent platform designed to support innovation at every layer, from experimentation to scale.

Our mission is to:

- **Empower developers** with modular tools to create powerful, observable AI agents
- **Make AI infrastructure safe**, transparent, and widely accessible
- **Monetise AI** contributions with open standards and a sustainable revenue model

MUXI's mission is to support the global transition to AI-native workflows by providing developers with everything they need to build intelligent agents: frameworks, tools, infrastructure, and economic incentives. Our vision is to make AI development as modular, transparent, and monetisable as app development is today.

3. Innovation

MUXI introduces significant technical and commercial innovations:

Technical innovation

- Declarative agent architecture that simplifies agent design
- Secure and observable execution (runtime logs, memory inspection, chain-of-thought tracking)
- Agent-to-agent communication (local and remote)
- Multiplexed architecture for handling multimodal and multichannel I/O
- Tooling (via MCP) to safely integrate agents with external systems

Business model innovation

- A Vercel-style cloud platform for agent orchestration and deployment
- A marketplace to enable monetisation and reuse of intelligent agents
- A community license allowing broad use while protecting hosted IP value

4. Problem space

AI adoption is rising, but agent-based workflows remain inaccessible due to:

- Lack of standardisation in agent architecture
- Limited observability and safety tools for AI execution
- Inability to monetise or distribute reusable agents
- Absence of open, trusted infrastructure for scalable AI adoption

MUXI addresses all of these with a complete solution stack that includes open-source components, hosted services, and economic incentives.

5. Product architecture

MUXI includes:

- **Framework:** An open-source, declarative-first runtime for building persistent, modular, and observable agents.
- **Tooling:** A set of reusable hosted services such as email, calendar, search, HTTP/webhooks, file storage, and more, usable by both self-hosted and cloud-hosted agents.
- **MUXI Cloud:** A production-ready, Vercel-like experience for hosting agent orchestrators, with agent logs, memory, sandboxing, billing, and multi-agent session management.
- **Marketplace:** A monetisation layer where developers can list, license, and monetise their agents or MCP modules.
- **Partner Layer:** Service providers and integrators who will build industry-specific workflows on top of MUXI's ecosystem.

MUXI supports agent-to-agent communication, memory tracking, and deterministic agent routing. This lets developers move beyond task-based LLM tools and into reusable, long-running software logic with shared context and secure orchestration.

MUXI is also built to support hybrid interaction: agents can interact with humans, APIs, other agents, and external services simultaneously.

MUXI is structured into five layers:

1. **MUXI Framework** – open-source declarative agent framework with observable runtime
2. **Tooling** – modular services for agents (e.g., calendar, email, search)
3. **MUXI Cloud** – hosted platform with memory, logs, billing, and orchestration
4. **Marketplace** – agent and tool discovery, licensing, and monetisation
5. **Partner Layer** – agencies and consultants that deliver custom solutions using MUXI

6. Go-to-market strategy

Phase 0: Framework development (pre-funded)

- Initial open-source release (currently in progress)
- Non-compete license: commercial use permitted except for hosting platforms

Phase 1: Community-first launch (Months 3–8)

- Developer content, documentation, templates, and SDKs
- Discord, GitHub presence, live meetups (EU/UK/US)
- Full-time DevRel and Community Manager to lead adoption

Phase 2: Hosted Tooling & MCP Servers (Months 6–8)

- Community-facing backend tools
- Emphasis on security, observability, and open-source extendability

Phase 3: MUXI Cloud (Months 9–12)

- Vercel-style self-service deployment platform
- Agent runtime orchestration, team support, logs, and abuse controls

Phase 4: Marketplace (Months 13–15)

- Developer monetisation for agents and tooling
- Revenue share, moderation, analytics, and payout infrastructure

Phase 5: Partner programme (Months 15–18)

- Integration partners, consultants, and agencies
- Sales and client success operations via MUXI Inc (US-based)

7. Hiring roadmap

Role	Source	Phase	Annual Cost
Lead Architect (Ran)	Grant	0–5	\$60,000
Developer Advocate	Grant	1	\$85,000
Community Managers (EU/US Split)	Grant	1	\$50,000
Evangelists (EU/UK/US Split)	Grant	2–4	\$100,000
Backend Devs (Automaze)	Grant	2–4	\$148,800
Frontend/Infra Devs	Grant	3	\$60,000
Partner Manager (EU/US)	Self-funded	5	\$80,000

8. Revenue model

MUXI will monetise through four channels:

8.1. MUXI Cloud

MUXI Cloud will offer a pricing structure optimised for individual developers, teams, and enterprises.

Example pricing structure:

Tier	Features	Monthly Price
Free	Shared infra, limited compute & logs	\$0
Developer	1 shared VM (1vCPU / 2GB), private orchestration	\$49
Pro	1 dedicated VM (2vCPU / 4GB), unlimited agents	\$119
Team	4x dedicated VM (8vCPU / 16GB), team features	\$499
Enterprise	BYO infra, SLAs, isolated cloud	Custom

Additional compute: \$200 per 4vCPU/8GB unit upgrade

8.2. Tooling and MCP Servers

Usage-based billing on high-volume shared services such as:

- Web search, scheduling, email parsing, scraping, HTTP clients
- Tiered or per-call pricing, depending on usage class

8.3. Marketplace

Agents and MCP modules will be offered in a curated developer marketplace with:

- 20% platform commission
- Licensing, usage gating, and optional approval processes

8.4. Custom Integrations

Organizations that prefer not to build agents in-house can hire Automaze to deliver tailored MUXI-based implementations, integrations, and automation workflows. This provides a scalable, service-based revenue stream to support businesses seeking ready-made solutions.

9. Financial projections

	OSS Users	Cloud Users	Revenue			
			Cloud	Tooling	Marketplace	Total
Q1	10,000	0	\$0	\$0	\$0	\$0
Q2	50,000	0	\$0	\$0	\$0	\$0
Q3	75,000	0	\$0	\$15K	\$0	\$15K
Q4	100,000	2,000	\$3M	\$250K	\$0	\$3.25M
Q6	500,000	15,000	\$20M	\$1M	\$500K	\$21.5M
Q8	1,000,000	50,000	\$60M	\$5M	\$10K	\$75M

10. Licensing strategy

MUXI uses an **open source licensing strategy** designed to encourage open adoption while protecting long-term sustainability.

The MUXI framework will be licensed using a permissive open source license (e.g., **Apache 2.0**) with a **single commercial caveat**: the license will prohibit launching MUXI-based **cloud-hosted platforms** that compete directly with the official MUXI Cloud.

This model:

- Encourages businesses to build commercial apps, services, and internal tools using MUXI
- Prevents direct cloning of the hosted orchestration layer ("Vercel for agents")
- Promotes OSS collaboration while protecting MUXI's cloud revenue model

Organizations will be free to:

- Self-host MUXI agents
- Extend or modify the framework for private or internal use
- Build commercial tools on top of MUXI

But may not:

- Build competing hosted infrastructure-as-a-service platforms based on MUXI's open-source core

11. Team

The MUXI team is intentionally lean by design. As a platform built to orchestrate intelligent agent workflows, we use our own internal agent swarm to handle a significant portion of planning, development, content creation, support, and operational coordination. This allows us to operate at high efficiency with a small core team, demonstrating the practical power of our own technology while maintaining agility and focus.

Leadership

Ran Aroussi, Founder & Architect, is a veteran developer, open-source creator, and product strategist with over 30 years of experience in building high-scale infrastructure and developer tooling. He is best known for creating the `yfinance` Python library (**used daily by millions of developers**) and for his early contributions to affiliate marketing automation, fintech platforms, and data-driven APIs.

Ran founded Automaze Ltd. to provide software development and CTO-as-a-Service to startups and SMEs. With deep experience in product-market fit, developer experience, and automation tooling, Ran brings both technical and entrepreneurial leadership to MUXI.

Core delivery team

Automaze Ltd (UK) leads the technical implementation with a flexible pool of senior developers hired per milestone. All engineering work, IP ownership, and technical R&D remain UK-based.

Community and commercial

Additional team members will include Developer Advocates, Community Managers, and Evangelists to support go-to-market adoption, community traction, and onboarding support. A US-based partnerships coordinator will be engaged in Phase 5 to handle integration opportunities.

High-level allocation of responsibilities

- **Automaze Ltd (UK)** will lead product and engineering delivery, providing experienced development and infrastructure support under the direction of the EU company.
- **Automaze Inc (US)**, a wholly owned US subsidiary may be formed at a later stage to manage go-to-market activities, enterprise sales, and ecosystem partnerships in North America.

12. Grant request

Total requested: **\$340,000**

- Phase 1: Community, ~\$140,000
- Phase 2: Tooling, ~\$40,000
- Phase 3: Cloud, ~\$160,000

13. Strategic fit

MUXI aligns with grant policy in:

- Driving AI adoption across public and private sectors
- Enabling safe and traceable AI workflows
- Supporting OSS-based exports and developer monetisation
- Creating high-value IP and jobs in the UK

14. Legal & Organizational Structure

This grant application is submitted by Ran Aroussi, an EU citizen and technical founder. A new EU-based SME will be formed immediately upon successful funding approval. The company will be fully owned and controlled by EU citizens and will hold all intellectual property and grant-related activities. Automaze Ltd (UK) will act as a contracted delivery partner under this structure.

14. Closing remarks

MUXI is positioned to become the foundational infrastructure for the AI agent ecosystem. With open-source roots, a scalable hosted platform, and a robust monetisation engine, we are ready to lead this new phase in AI adoption.

We welcome grant partnerships, pilot collaborations, and strategic alliances.

15. Contact

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APPENDIX A

What are AI Agents?

AI agents are autonomous software programs capable of perceiving information, making decisions, and executing tasks across digital systems. They combine language models (LLMs), memory, and external tools to simulate intelligent reasoning and behaviour.

Think of an AI agent as a **digital assistant** that doesn't just respond to commands – it can read, understand, and take initiative. It can autonomously coordinate tasks, interact with APIs, schedule meetings, draft content, and even collaborate with other agents. Unlike simple automations or chatbots, AI agents are **context-aware**, modular, and extensible.

Examples include:

- **Chatbots:** Handling conversations and queries.
- **Virtual assistants:** Managing tasks like scheduling or controlling devices.
- **Autonomous systems:** Self-driving cars or robotic agents navigating physical spaces.
- **Recommendation systems:** Algorithms suggesting content on platforms like Netflix.

How agents work (under the hood)

Today, most AI agents operate using a combination of large language models (LLMs) and structured "function calling." A user (or another system) sends input to an agent, which is processed by the LLM. The model identifies the user's intent, determines which tools (or "functions") to use, and calls those tools in a sequence. These functions often include tasks like fetching emails, querying databases, generating summaries, or interacting with APIs.

This process typically involves:

1. **Prompt processing** – User input is wrapped in a structured prompt.
2. **LLM parsing** – The LLM interprets the prompt and returns a response with tool calls.
3. **Function execution** – External code executes those tools (e.g., via OpenAI tool-calling or LangChain/AutoGen APIs).
4. **Response generation** – The agent wraps up the results and delivers a final output.

Agents often run as a series of stateless loops, making them hard to scale, chain, or monitor without additional architecture.

How MUXI simplifies agent development

MUXI simplifies the complexity of building and running AI agents. Developers can define agents declaratively using YAML or JSON, or fully programmatically with code. The MUXI framework handles orchestration, tool invocation, memory management, and agent-to-agent messaging – freeing developers from having to build all these layers themselves.

MUXI abstracts the complexity of building these pipelines by introducing a declarative and composable framework that manages:

- **Prompt engineering**
- **Function registration** (via MCP tools)
- **Memory persistence and replay**
- **Logging and observability**
- **Execution orchestration**

Whether you're running locally or on the cloud, MUXI gives developers a robust runtime to safely deploy intelligent agents without reinventing the wheel.

A simple flowchart showing:

Input → Context + Memory → LLM + Logic Layer → Tool Execution (via MCP) → Output

Below the diagram, optional labels:

- “Input sources: user, webhook, file, email”
- “LLM reasoning + memory injection”
- “MCP tools: email, scraping, calendar, search, etc.”
- “Output: reply, action, delegation”

This helps non-technical readers visualise how agents process and respond in real time.

APPENDIX B

Sector-specific use cases

Public Sector: NHS Clinical Efficiency Agents

AI agents can reduce admin time and increase efficiency in frontline clinical environments. For example:

- An agent running **locally inside a clinic** can **record patient consultations**, transcribe them using **on-premises LLMs**, and summarise the interaction for clinical records.
- This ensures **data privacy** (no external services used) and frees up doctors to focus on patients instead of documentation.
- Additional agents can triage referrals, prepare shift handovers, and automate briefings.

These interventions could **significantly reduce NHS waitlists**, improve patient throughput, and reduce clinician burnout.

Private Sector: Financial Compliance Agent for SMEs

SMEs struggle with tax filings, VAT returns, and compliance paperwork. A MUXI agent can:

- Sync with Xero, QuickBooks, or spreadsheets
- Draft reports, flag issues, and email reminders
- Reduce reliance on expensive manual processes

By automating compliance for millions of UK businesses, this type of agent could meaningfully support SME productivity, reduce tax delays, and improve financial hygiene across sectors.

APPENDIX C

About Automaze, Ltd.

Automaze Ltd is a UK-based software engineering agency founded and led by Ran Aroussi. The company specialises in building infrastructure and automation systems for startups, SMEs, and product-led teams.

Automaze provides:

- CTO-as-a-service and fractional technical leadership
- High-scale backend and infrastructure development
- End-to-end product delivery: from planning and architecture to DevOps and post-launch scaling

The company has helped dozens of businesses – from early-stage founders to enterprise product teams – launch modern digital products. Its ethos blends technical excellence with founder empathy: helping innovators move faster, reduce waste, and build with confidence.

Automaze is the core development partner for MUXI and will lead all platform implementation, infrastructure delivery, and roadmap execution.

APPENDIX D

MUXI Framework Whitepaper

We've released a whitepaper, describing MUXI Framework – the underline technology for this project.

You can find it in PDF format here:

“[MUXI](#): A modular, open-source framework for building multi-AI-agent systems orchestration with persistent memory, standardized communication protocols, and chain-of-thought tracing.”

<https://automaze.io/muxi-whitepaper>